

Introduction to Machine Learning



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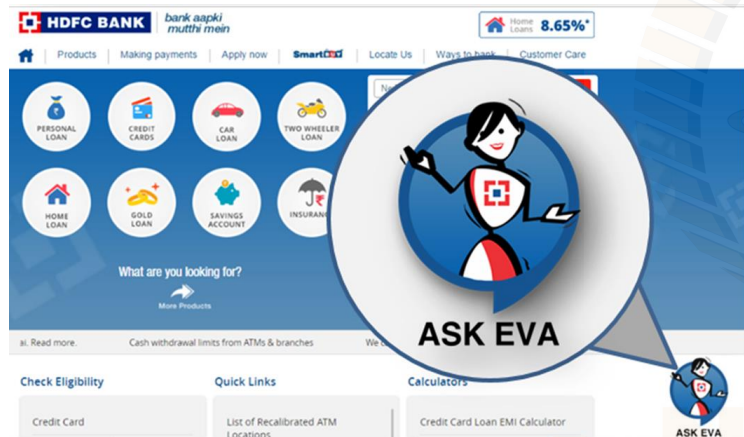
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Introduction to Machine Learning



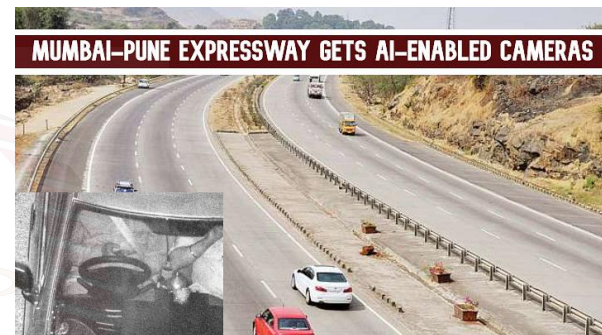
Motivation

Some real-world examples of ML around us...



AI-powered chatbot for customer query handling

<https://timesofindia.indiatimes.com/business/india-business/hdfc-bank-launches-artificial-intelligence-driven-chatbot-eva/articleshow/57481168.cms>



Traffic Policing & Challan

<https://www.91wheels.com/news/indias-first-ai-enabled-cameras-installed-on-mumbai-pune-expressway>

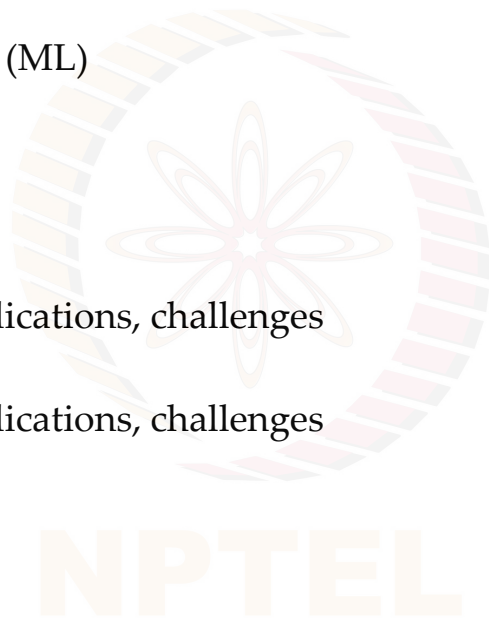


Sana, a virtual news anchor (Aaj Tak)

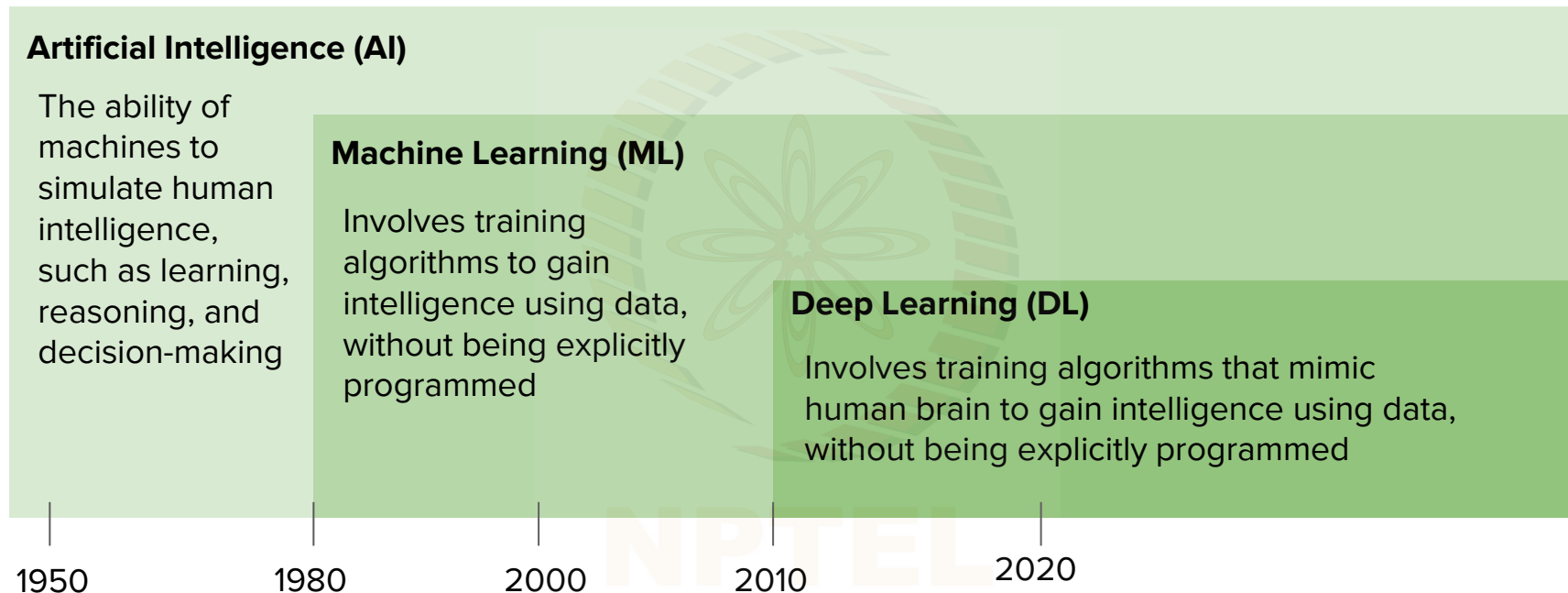
<https://www.jivanhindi.in/2023/04/ai-anchor-sana-kaun-hai-in-hindi.html>

Learning Objectives

- Understand Machine Learning (ML)
 - Past & present
 - Definition
 - Types
- Supervised ML
 - Definition, example, applications, challenges
- Unsupervised ML
 - Definition, example, applications, challenges
- Applications
- Practical Aspects



Evolution of AI over time



ML is a subset of AI and a power set of DL.

History of Machine Learning

- Arthur Samuel, an American AI-pioneer, is credited to popularize the term *Machine Learning* (1959) [1], who explained ML as,

“Field of study that give computers the ability to learn without being explicitly programmed”

- The 1980s
 - emergence of neural networks (NN) technique
 - NN limitations
 - Hindered to widespread adoption: data sparsity & low computing capabilities
 - First AI-Winter ([10.1109/MIS.2008.20](https://blog.hnf.de/als-ein-computer-dame-lernte/))



Samuel's checkers-playing program was among the world's first successful self-learning programs.

<https://blog.hnf.de/als-ein-computer-dame-lernte/>

[1] A. L. Samuel, "Some Studies in Machine Learning Using the Game of Checkers," in *IBM Journal of Research and Development*, vol. 3, no. 3, pp. 210-229, July 1959, doi: 10.1147/rd.33.0210.

Machine Learning Today

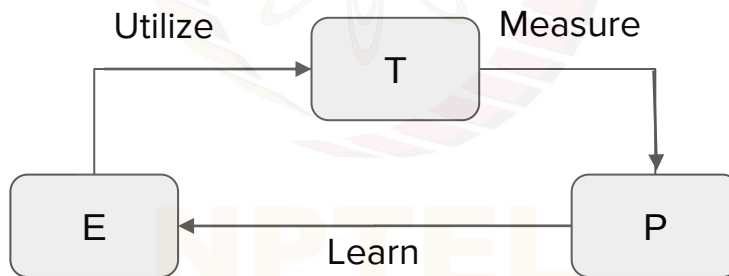
- The 2000s
 - development of support vector machines (SVM) & decision trees
 - able to handle larger datasets
 - more computationally efficient [2]
 - rise of big data
 - development of deep learning (DL)
- Today,
 - ML's wide-spread applications - from image recognition and natural language processing (NLP) to self-driving cars and personalized medicine.
 - DeepFace: DL-based Facial Recognition
 - AlphaGo: AI-player of 'Go' board game [defeated world champion]
 - ChatGPT: AI-chatbot, and so on...



The Three Godfathers of AI: Yann LeCun, Geoffrey Hinton, and Yoshua Bengio are known for their pioneering contributions in the development of AI <https://fortune.com/2019/04/02/eye-on-ai-godfathers-deep-learning/>

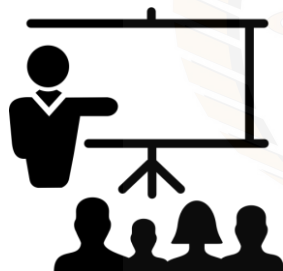
Definition

“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E [3].”



Types

- Supervised Learning
- Unsupervised Learning



Supervised Learning

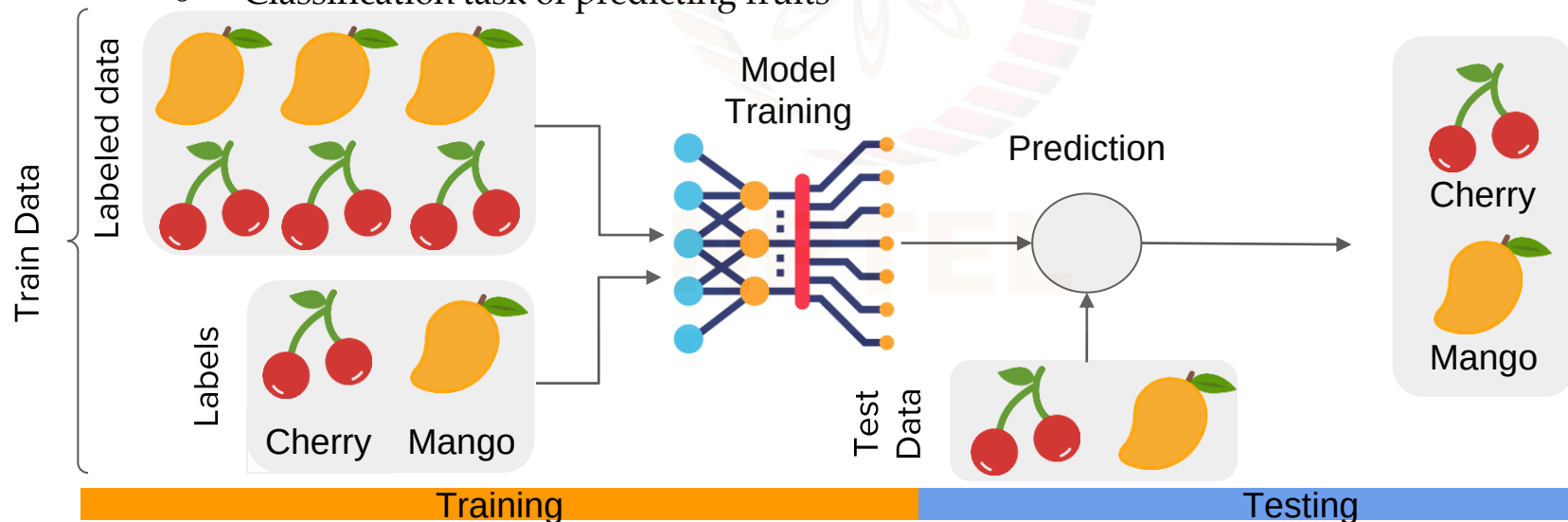


Unsupervised Learning

Supervised Machine Learning

- Algorithm learns to map input data (features) to known output data (labels).
- Training data is labeled: each data point has a corresponding output label that the algorithm is trying to predict.
- Tasks:
 - **classification**: the output variable is a categorical variable (response is qualitative)
 - **regression**: the output variable is a continuous variable (response is quantitative)
- Example:
 - Classification task of predicting fruits

Class = 0
Cherry
Class = 1
Mango



Supervised Machine Learning

- Some Applications:
 - **Image classification:** Identify objects or features in images, such as classifying whether an image contains a cat or a dog.
 - **Sentiment analysis:** Determine the sentiment or opinion expressed in text data, such as classifying whether a product review is positive or negative.
 - **Fraud detection:** Identify fraudulent transactions in financial data, such as predicting whether a credit card transaction is likely to be fraudulent.
 - **Predictive maintenance:** Predict when maintenance is needed for equipment based on sensor data, such as predicting when a machine is likely to fail.
 - **Personalized recommendations:** Recommend products or content to users based on their past behavior and preferences, such as suggesting movies or TV shows to watch.



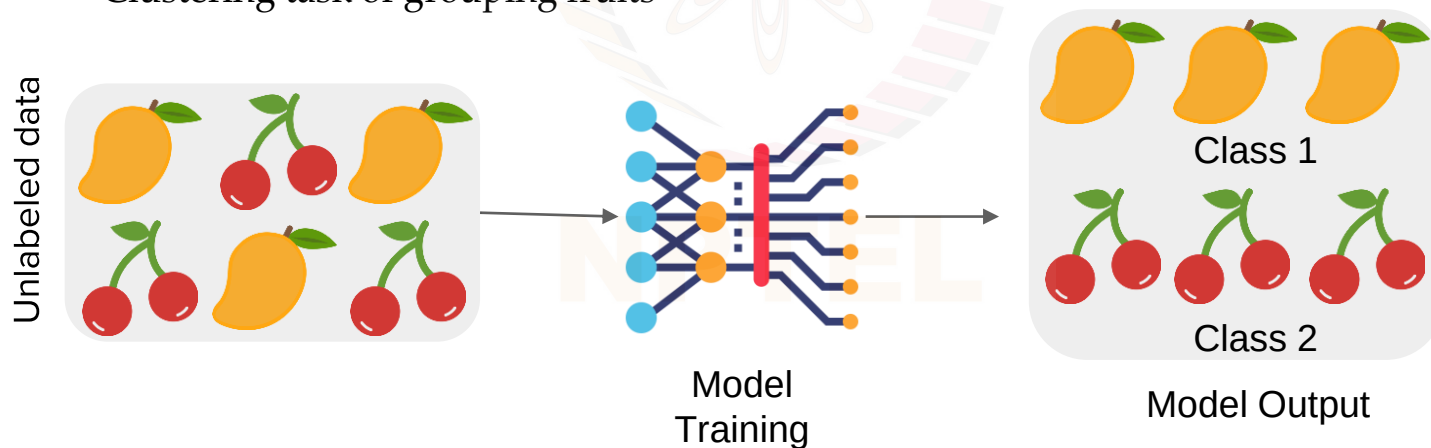
Supervised Machine Learning

- Challenges:
 - **Insufficient or biased data:** Lack of data or biased data can lead to poor model performance or inaccurate predictions.
 - **Overfitting:** When the model is too complex, it can fit the training data too closely, leading to poor generalization performance on new data.
 - **Feature engineering:** Choosing and engineering the right set of features can be a time-consuming and subjective process that requires domain expertise.
 - **Model selection:** Choosing the right model and hyperparameters for a given problem can be challenging and require extensive experimentation and tuning.
 - **Interpretability:** Understanding why a model makes certain predictions or decisions can be difficult, especially for complex models like deep neural networks.



Unsupervised Machine Learning

- Algorithm learns patterns and structures in the input data (features) without being given explicit labels / targets.
- Training data is unlabeled.
- Tasks:
 - **clustering**: group similar data points together
 - **dimensionality reduction**: the output represents the input with reduced dimensions
- Example:
 - Clustering task of grouping fruits



Unsupervised Machine Learning

- Applications:
 - **Anomaly detection:** Identify unusual or rare events or patterns in data, such as detecting fraudulent transactions.
 - **Clustering:** Group similar data points together based on similarity or distance, such as clustering customers based on their purchasing behavior.
 - **Dimensionality reduction:** Reduce the number of features or variables in high-dimensional data while preserving as much information as possible, such as reducing the number of dimensions in an image or text dataset.
 - **Topic modeling:** Discover underlying topics or themes in a collection of documents or text data, such as identifying topics in customer reviews or news articles.



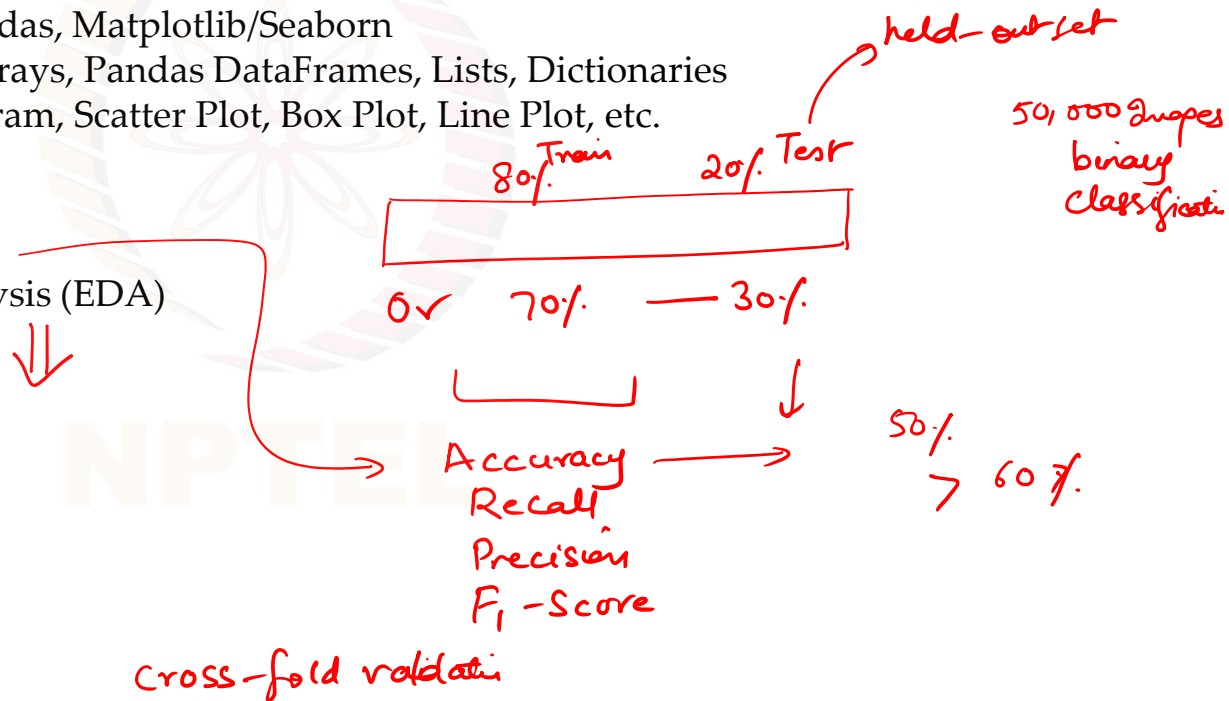
Unsupervised Machine Learning

- **Challenges:**
 - **Evaluation:** There are no clear evaluation metrics for unsupervised learning, making it difficult to objectively compare different models or algorithms.
 - **Interpretability:** Understanding the meaning or interpretation of the patterns or clusters discovered by unsupervised learning can be challenging, especially for complex models or high-dimensional data.
 - **Scalability:** Some unsupervised learning algorithms can be computationally expensive and difficult to scale to large datasets or high-dimensional data.
 - **Determining the number of clusters:** Determining the optimal number of clusters in a clustering algorithm can be challenging and subjective.
 - **Data preprocessing:** Preprocessing and cleaning the data can be time-consuming and require domain expertise, especially for complex or unstructured data like images or text.



Practical Aspect: Tutorials

- Tutorials [Week 2 onwards] on Google Colab (Any other interface you are comfortable with, is acceptable)
- Expected Basic Python. Should be comfortable with
 - **Libraries:** Numpy, Pandas, Matplotlib/Seaborn
 - **Data types:** Numpy Arrays, Pandas DataFrames, Lists, Dictionaries
 - **Visualizations:** Histogram, Scatter Plot, Box Plot, Line Plot, etc.
- Concepts:
 - Train-Test Split
 - Evaluation Metrics
 - Exploratory Data Analysis (EDA)

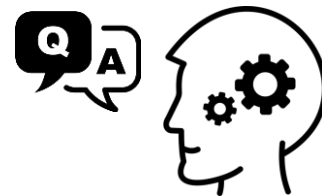


Let us Try!

Question: Which of the following is an example of unsupervised machine learning?

- A) Predicting stock prices
- B) Recognizing handwritten digits
- C) Clustering customer segments
- D) Identifying spam emails

You may pause the video and try.



Let us Try!

Question: Which of the following is an example of unsupervised machine learning?

- A) Predicting stock prices
- B) Recognizing handwritten digits
- C) Clustering customer segments
- D) Identifying spam emails

Explanation:

Predicting stock prices (A) and recognizing handwritten digits (B) are examples of supervised machine learning tasks, where the algorithm learns to predict an output variable based on known ground truth of the training samples. Identifying spam emails (D) is also a supervised machine learning task. Clustering customer segments (C), on the other hand, is an example of unsupervised machine learning, where the algorithm learns to group similar training data samples together based on their features, without being explicitly told what the correct output groups should be.

To Summarize

What we learnt in this module is as follows:

- Formal introduction to ML
- Supervised ML
- Unsupervised ML
- Practical Aspects
- Solved a question

